

RESIN 3D PRINT GUIDE

VORATH MINIATURES • CALIBRATION, SUPPORTS, WASHING & CURING

PRODUCTION GUIDE V1.0

Universal Standard: VORATH miniatures are sculpted in 28–32mm heroic tabletop scale. To preserve the sharp faceting of crystal blades, halos, and sacred-geometry sigils, follow this calibration standard for mSLA / LCD resin 3D printers.

1. LAYER HEIGHT & SLICING ORIENTATION

- Optimal Layer Height: 0.03mm to 0.05mm** (0.03mm produces nearly invisible layer lines on crystal faceting).
- Tilt Angle:** Orient figures tilted **35° to 45° backwards** relative to the build plate.
- Face Preservation:** Never orient the miniature facing directly towards the build plate. Placing supports along the back, cloak folds, and heels keeps facial masks and chest sigils pristine.
- Suction Reduction:** Never print large cloaks or bases parallel to the FEP film. Angle them to slice with a continuous, gradual cross-sectional peel area.

2. EXPOSURE BENCHMARKS (STANDARD 8K / 12K LCD)

RESIN TYPE	LAYER HEIGHT	NORMAL EXP.	BURN-IN (BOTTOM)
Standard 8K Resin	0.05mm	1.8s – 2.4s	25s – 30s (5 layers)
ABS-Like / Tough Resin	0.05mm	2.2s – 2.8s	30s – 35s (6 layers)
High Detail (Fine 0.03mm)	0.03mm	1.4s – 1.9s	22s – 28s (5 layers)

**Temperatures below 20°C (68°F) increase resin viscosity. Increase normal exposure by 10–15% in cold rooms.*

3. SUPPORT SIZING & PLACEMENT STRATEGY

- Heavy Anchors (0.60mm tip):** Place on the lowest initial islands (feet, soles, hemlines) to securely anchor the peel force.
- Medium Supports (0.35mm–0.40mm tip):** Torso, weapons, thick limbs, and main structural overhangs.
- Light Supports (0.20mm–0.25mm tip):** Delicate crystal tips, censer chains, fingers, and halos.
- Raft Type:** Use a skate / chamfered raft with 1.2mm–1.5mm thickness for clean removal from the build plate without chipping.

4. THE WARM-WATER SUPPORT REMOVAL SECRET

Do NOT remove supports after curing! Remove supports immediately after washing, while the resin is still green:

- Submerge the washed miniature into a bowl of **warm water (~50°C / 120°F)** for 45–60 seconds.
- The heat temporarily softens the thin support contact tips while keeping the model rigid.
- Peel the supports away like zipper teeth. Tips snap off flush with **zero surface pitting, divots, or broken blades**.
- Use flush-cutters for heavy connection nubs along the underside of the feet.

5. TWO-STAGE WASHING PROTOCOL

- Stage 1 (Dirty Wash):** 2–3 minutes in 99% Isopropyl Alcohol (IPA) to strip bulk uncured resin.
- Stage 2 (Clean Wash):** 2 minutes in clean, fresh IPA bath (or ultrasonic cleaner).
- Crucial Step:** Allow the miniature to **dry completely (15–20 minutes)** before post-curing. Exposing damp alcohol to UV light creates an irreversible cloudy white crust.

6. UV POST-CURING & PAINTING PREP

- UV Cure Time: 2 to 4 minutes** in a 405nm curing station. Turn the model midway through.
- Avoid Over-Curing:** Curing beyond 5–6 minutes increases brittleness in delicate weapons and crystal points.
- Primer Recommendations:** Smooth matte black primer with a white or grey zenithal highlight brings out the faceted crystal geometry perfectly.

7. SCALE & TABLETOP COMPATIBILITY

Models are natively scaled for **28–32mm heroic skirmish wargaming**:

- Infantry (Revelers, Justiciars, Mutants): Standard 25mm–32mm circular/hex bases.
- Heroes (The Prism Saint): 32mm–40mm scenic base.
- Centerpiece (Abyssal Ouroboros): 50mm–60mm oval/round base.